

High-Efficient Rectifier With Wide Dynamic Range Using Self-Bias Junction Capacitance Compensation

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Abstract—This letter presents a novel rectifier assisted by a self-bias junction capacitance compensation (JCC) for high-efficiency RF-to-dc conversion across a wide input power range. Unlike conventional tunable matching networks at the input, the proposed approach dynamically turns on the JCC in parallel with the rectifying diode by sampling the output voltage. This compensation mitigates intrinsic junction capacitance drift caused by the input power, significantly enhancing impedance stability without introducing extra matching circuit or requiring external bias power. The power consumption of the proposed JCC is less than 0.2% of the output dc power, making it negligible. Measured results show that the power conversion efficiency (PCE) remains above 50% over the input power range of -2 to 24 dBm, and exceeds 70% between 8 and 22 dBm, with a peak efficiency of 81.5% at 21 dBm. Compared to a reference design with a fixed impedance matching, the proposed JCC significantly broadens the ranges of input power and dc load at the same time.

Index Terms—Diode, junction capacitance compensation (JCC), RF rectifier, wide input power range, wireless power transmission (WPT).

I. INTRODUCTION

WITH the rapid development of the fifth generation (5G) communication and Internet of Things (IoT) devices, wireless energy harvesting (EH) and wireless power transmission (WPT) have attracted significant attention [1], [2]. The rectifier, which converts RF energy into dc output, is a key device in EH/WPT systems and has been extensively studied. However, in real-world applications, a rectifier's available RF energy and load are generally not constant but vary with operating conditions. For instance, the input power of the rectenna deployed on a space-based system or an aircraft has a considerable fluctuation [3], [4]. The fluctuating input power and output load result in impedance mismatching due to the

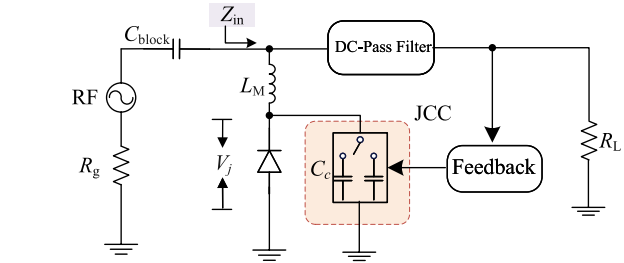


Fig. 1. Block diagram of the proposed rectifier with a JCC circuit biased by the output voltage. The JCC is adaptively switched in parallel with the diode when the input power or dc load is in the high region, then the intrinsic junction capacitance drift caused by the input power can be neutralized.

rectifying device's nonlinearity. Therefore, as a key component of WPT systems, it is crucial to design rectifiers with a wide range of output load and input power [5].

As pointed out in [6], most rectifiers achieve peak efficiency only at a specific input power range, because their matching networks are designed for a fixed input impedance. Several approaches have been proposed to extend the input power range in recent years. The first is to increase the breakdown voltage of the rectifying diode by a bypass switch such as pHEMT or MOSEFT. For instance, FET-assisted diode rectifiers have been demonstrated in [7], but this topology suffers from low operating frequency. The second approach introduces a power recycling network. In [8], the power reflected from the two main branches can be efficiently transmitted to the recycling branch by using a branch-line coupler. It is noted that such a coupler requires additional circuit area and inherently introduces some degree of insertion loss. The third one is impedance or resistance compression networks (RCNs) reported in [9] and [10]. Although the RCN lowers the sensitivity of the rectifier to the input power or the load, rectifiers based on RCNs require two sub-rectifiers, making the diode switching loss more evident in the low input power region.

Some newly published solution includes a matching network using varactor diodes [11] and adaptive input power distribution between a high-power and a low-power rectifier [12], [13]. Those approaches are effective, but close attention should be paid to the insertion loss of these introduced components. In this letter, considering the power-dependent impedance characteristic of the diode, we introduce a direct junction capacitance compensation (JCC) circuit in parallel with the diode, as shown in Fig. 1. The proposed JCC operates as a

Received 15 October 2025; accepted 10 December 2025. Date of publication 23 December 2025; date of current version 9 April 2026. This work was supported in part by NSFC under Grant 62101366 and Grant U22A2015 and in part by Wenzhou Scientific Research Project under Grant G2023073. (Corresponding authors: Changjun Liu; Pengde Wu.)

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Digital Object Identifier 10.1109/LMWT.2025.3644372

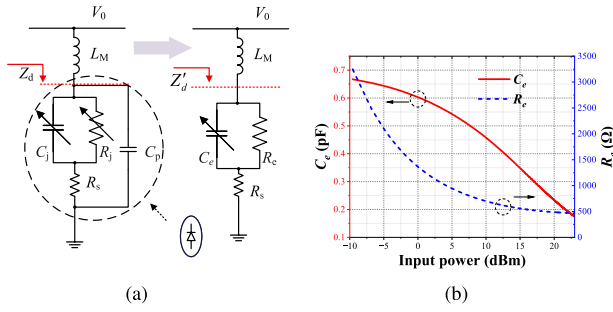


Fig. 2. (a) Detailed and simplified equivalent circuit of a diode. (b) Calculated C_e and R_e versus input power for the HSMS-282 diode.

self-biasing device that requires no additional logic control, automatically turning on/off according to the output voltage of the rectifier. The proposed JCC enhances impedance stability over input power and dc load, which enables a robust and high-efficiency rectifier design.

II. PRINCIPLE AND CIRCUIT DESIGN

A. Input Impedance Without JCC

The nonlinearity of Schottky diodes in EH circuits causes their impedance to vary significantly with the input power. Fig. 2(a) shows an equivalent circuit model for a shunt diode rectifier where L_M is utilized for compensating the capacitive impedance of the rectifying diode. As shown, V_0 is the produced output dc voltage across the resistive load R_L , while this voltage also biases the junction capacitance C_j at the same time. C_j decreases with V_0 , according to

$$C_j = \frac{C_0}{\left(1 + \frac{V_0}{V_{bi}}\right)^M} \quad (1)$$

where C_0 is the zero-bias junction capacitance, V_{bi} is the built-in potential, and M is the grading coefficient [14]. In an RF rectifier, V_0 equals to $(\eta_{rec} P_{in} R_L)^{1/2}$ where η_{rec} is the RF-dc conversion efficiency, then C_j decreases steadily as P_{in} increases according to (1).

In [15], closed-form equations are derived for the diode input impedance, which is denoted as Z_d . As shown in the right part of Fig. 2(a), Z_d is also equivalent to the parallel connection of a nonlinear junction capacitance C_e and a nonlinear junction resistance R_e , and its impedance is

$$Z'_d = \frac{1}{\frac{1}{R_e} + \frac{1}{j\omega C_e}}. \quad (2)$$

Now let $Z_d = Z'_d$ give

$$C_e = C_j \frac{\pi - \theta_{on} + \frac{1}{2} \sin 2\theta_{on}}{\pi}; \quad R_e = \frac{\pi R_s}{\theta_{on} - \sin \theta_{on} \cos \theta_{on}} \quad (3)$$

where θ_{on} is the forward-bias turn-on angle. θ_{on} is a dynamic variable dependent on V_0 and R_L , it is expressed as

$$\frac{\pi R_s}{R_L \left(1 + \frac{V_{bi}}{V_0}\right)} = \tan \theta_{on} - \theta_{on} \quad (4)$$

where V_{bi} is the diode's built-in voltage [16].

The increase in input power leads to an increase in V_0 and θ_{on} , which consequently reduces C_e and R_e according to (3).

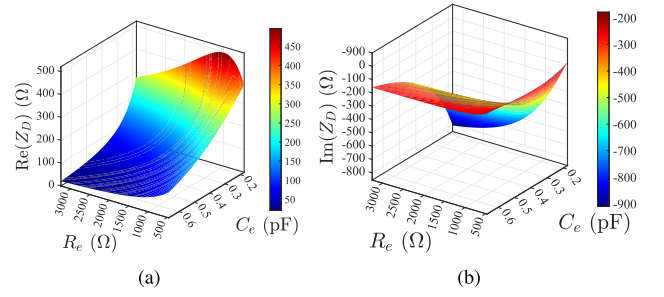


Fig. 3. Calculated (a) real part and (b) imaginary part of diode impedance using the values of C_e and R_e from Fig. 2(b).

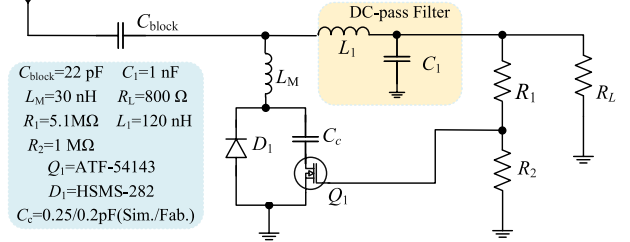


Fig. 4. Schematic and circuit parameters of the rectifier with wide input power dynamic range. The proposed JCC employs an N -channel FET in series with a capacitor C_e (0.25 pF for simulation, 0.2 pF for fabrication). The sum of R_1 and R_2 is ~ 2000 times larger than the dc load R_L , so the power consumption of JCC is negligible.

Fig. 2(b) presents the calculated C_e and R_e versus RF input power for a single-shunt diode RF rectifier. These results correspond to a single-shunt diode RF rectifier utilizing an HSMS-282B diode, with a dc load of 800 Ω and an operating frequency of 915 MHz. Starting at around 0.67 pF at -10 dBm, C_e (red curve) decreases monotonically with increasing RF input power before reaching 0.23 pF at 20 dBm. While R_e decreases steadily with the input power.

With the equivalent circuit described by C_e and R_e , the diode input impedance Z'_d is expressed as

$$Z'_d = R_s + \frac{R_e}{1 + (R_e C_e \omega)^2} - j \frac{R_e^2 \omega}{1/C_e + C_e (R_e \omega)^2} \quad (5)$$

where ω is the angular frequency.

As can be seen in (5), the real part of Z'_d decreases monotonically as the C_e increases. While the denominator of the third term of (5) increases with C_e when $C_e > 1/(R_e \omega)$, which leads to a decrease of the imaginary part of Z'_d . To have a better understanding, Fig. 3 presents the calculated Z'_d using the values of C_e and R_e from Fig. 2(b). It is clear that the real part of the diode impedance decreases as the C_e increases, and C_e has a greater weight than R_e .

Based on the results shown in Fig. 3, C_e is mainly responsible for the impedance variation of the diode impedance. In this letter, to naturalize the impedance variation caused by C_e , as shown in Fig. 1, direct compensation to the junction capacitance is proposed. Since junction capacitance decreases with the increase of input power, a JCC for compensation is adaptively switched in parallel with the diode when the input power is high.

B. Input Impedance With the Proposed JCC

As shown in Fig. 4, the proposed JCC is parallel with the rectifying diode, which employs an N -channel FET

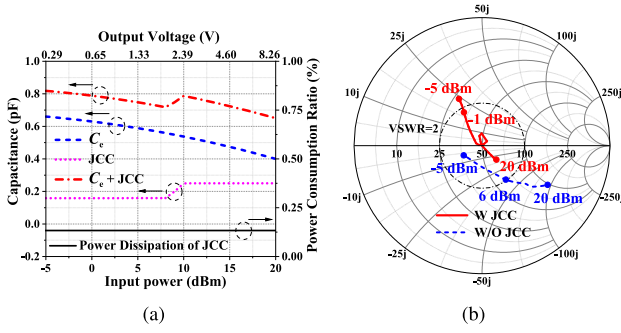


Fig. 5. (a) Simulated capacitance and dc power consumption of the proposed JCC. (b) Improved impedance matching with a wide range of input power by the proposed JCC.

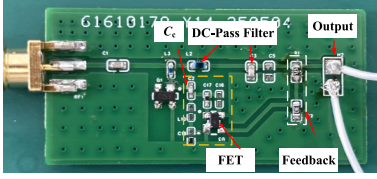


Fig. 6. Fabricated rectifier incorporating the proposed JCC.

(ATF54143) in series with a capacitor C_c . The switching of the FET is controlled by a voltage divider (R_1 and R_2) which samples the output voltage, and it is switched “on” when the input power is high and V_0 is above a threshold voltage. This configuration effectively mitigates junction capacitance variations induced by the input power, ensuring impedance matching within a wide range of input power.

Fig. 5(a) presents the simulated capacitance and power consumption of JCC versus input power. The compensation by JCC is dynamically switched on around 8 dBm, naturalizing for the inherent junction capacitance variation of the diode. More importantly, the simulated dc power consumption of JCC is less than 0.2% of the output dc power, which is negligible.

Finally, Fig. 5(b) demonstrates the improved impedance matching with dynamic compensation by the proposed JCC. Compared with the results without using JCC, the trajectory of Z_{in} over is curved, showing a good matching ability within a larger input power range (say: -1 to 20 dBm).

III. IMPLEMENTATION AND MEASUREMENT

To validate the proposed JCC for input power range extension, a rectifier prototype operating at 915 MHz was manufactured with FR-4 substrate. A photograph of the fabricated rectifier is shown in Fig. 6 according to the schematic and component parameters depicted in Fig. 4. C_c in the simulation is 0.25 pF, and it is set to 0.2 pF in the fabrication considering the parasitic capacitance. In the experiments, the rectifier was measured in terms of PCE and $|S_{11}|$ versus the input power, and comparisons to fixed compensations were also included. In the reference group where JCC is not used, FET is removed, and the capacitor C_c is connected to ground at one end.

In Fig. 7(a), measured $|S_{11}|$ versus input power for the rectifier with/without JCC are plotted. As shown, the JCC-based rectifier maintains impedance matching within a wide range of input power levels. The $|S_{11}|$ remains well below -10 dB from -9 to 24 dBm, demonstrating improved stability and efficiency compared to reference designs with fixed compensation. For

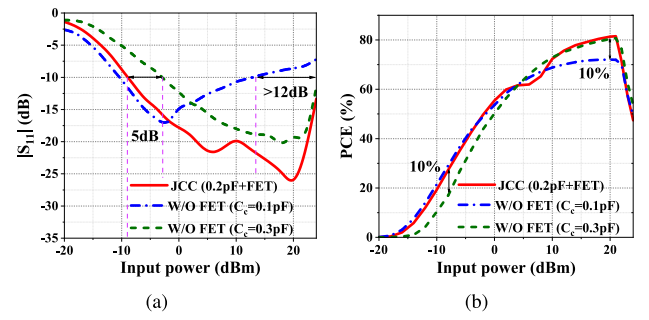


Fig. 7. (a) Measured $|S_{11}|$ and (b) PCE of rectifiers with and without JCC. In the reference group where JCC is not used, FET is removed, and C_c is connected to ground at one end.

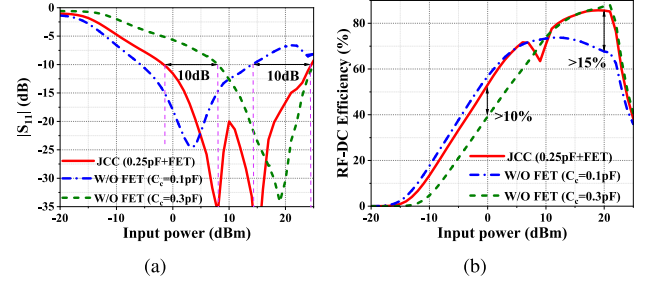


Fig. 8. (a) Simulated $|S_{11}|$ and (b) PCE of rectifiers with and without JCC.

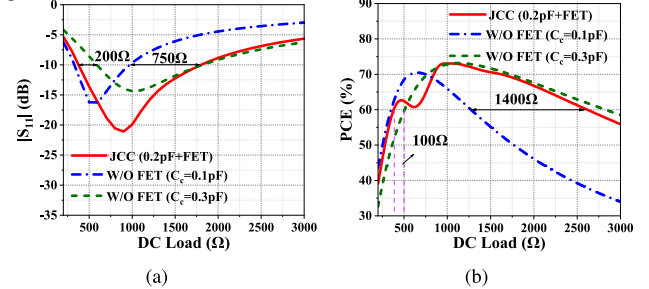


Fig. 9. (a) Measured $|S_{11}|$ and (b) PCE versus dc load with/without JCC, and the input power is 10 dBm.

example, in contrast to the green dashed line of 0.3 pF, the proposed JCC extends the input power range by 5 dB.

Fig. 7(b) depicts the measured PCE versus input power at 915 MHz. The JCC-based rectifier exhibits a 10% efficiency improvement over designs with fixed compensation, particularly at both low and high input power levels. The measured PCE is more than 50% within the input power range from -2 to 24 dBm, and a peak efficiency of 81.5% occurs at 21 dBm. It is observed that PCE has a local drop around 8 dBm, which is because the FET is in the ohmic region.

Fig. 7 validates the simulation results from Fig. 8, confirming that the proposed JCC significantly improves both impedance matching and efficiency across a wide input power range. The measured RF-dc efficiency is slightly lower than the simulation, but it shows strong agreement with the simulation, establishing the effectiveness of this design.

Fig. 9 presents the measured $|S_{11}|$ and PCE versus R_L at 10 dBm, respectively. Clearly, the proposed JCC extends the matched load range, achieving $|S_{11}| \leq -10$ dB over 0.4–1.8 k Ω wider than the two fixed matching designs. As shown in Fig. 9(a), the PCE follows the blue dashed-dotted line at first, then it switches to the green dashed line, enabling high efficiency in both low and high load segments. As R_L

TABLE I

COMPARISON WITH THE PRIOR RECTIFIERS FOR WIDE DYNAMIC RANGE

Ref	[17]	[18]	[19]	This work
Pow. range ($\eta > 50\%$)	23.3 dB	28 dB	22 dB	26 dB
Pow. range ($\eta > 70\%$)	6.5 dB	0 dB	8 dB	14 dB
Pow. range $ S_{11} \leq -10$ dB	N.A	40 dB	32 dB	33 dB
Peak Eff.	74.5% @ 27 dBm	68% @ 23 dBm	74.2% @ 10 dBm	81.5% @ 21 dBm
Fre. (GHz)	2.45	0.915	0.915	0.915
Topology	High& Low pow. branch	High& Low pow. branch	RPCN*	JCC
Diode: (HSMS-)	282×2 282×1	282×2 285×2	285×12	282×1
No. of branches	2	2	2	1

RPCN*: Reflected Power Compensation Network

increases, the PCE of the rectifier with JCC remains over 60% with the output load from 0.4 to 2.6 k Ω , achieving up to 175% improvement compared to that of a fixed 0.1 pF (0.4–1.2 k Ω).

Table I shows a comparison of the performances between the proposed design and the recently reported rectifier with a wide range of input power. It is noted that the proposed rectifier demonstrates the widest input power range for efficiency above 70% in existing literature, which is attributed to its low switching loss enabled by a single-diode configuration.

IV. CONCLUSION

To address impedance mismatching induced by input power variation, this work presents a self-bias JCC for a high-efficiency microwave rectifier. The proposed JCC introduces an adaptive compensation for diode matching at both low and high input power levels, naturalizing the impedance variation with the input power or dc load. Moreover, the power consumption of JCC is negligible because the sampling resistors for JCC are ~ 2000 times larger than the dc load R_L . The JCC's self-biasing architecture, powered by the rectifier output voltage, inherently eliminates the requirement for additional control logic circuit.

Experimental validation demonstrates that the rectifier exhibits more than 10% efficiency improvement over designs with fixed compensation, particularly at both low and high input power levels. In addition, the proposed JCC can also broaden the range of output dc load. In contrast to prior approaches—such as [20], which uses an FET to deactivate the low-power branch under high input power, and [21], which employs an FET to bypass a series diode stack to boost the rectifier's effective breakdown voltage—the proposed JCC directly compensates the diode's capacitance for stabilizing the diode impedance over a wide power range.

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